Abstract

We show that three of the deepest observational puzzles in modern cosmology—the cosmological constant fine-tuning problem, the 4–6σ Hubble tension, and the absence of WIMP dark matter signals—share a common structural origin within a single causal-ontological framework. The unifying principle is the *actualization density* $\lambda(x)$: the local rate at which irreversible causal interactions write permanent vertices into the spacetime graph. First, the vacuum contribution to the metric source is structurally zero—not fine-tuned to zero—because virtual (off-shell) processes produce no actualization events and therefore no metric sourcing. Second, the effective Hubble rate satisfies $H_{\text{eff}}(x) \propto 1/\lambda(x)$: early-universe probes and late-universe probes sample different actualization regimes, producing a real physical discrepancy rather than a systematic error. Third, causally silent particles (WIMPs) cannot gravitate because they produce no actualization flux; the dark matter phenomenology is explained instead by a Laplacian correction $\rho_\lambda \propto \nabla^2(\ln \lambda)$ in the metric sourcing equation. We derive the bandwidth-partition formula $H_{\text{local}}/\bar{H} = 1 - \delta\,(\bar{\lambda}_m/\bar{\lambda}_{\text{exp}})$, calibrate one free parameter from the KBC void, and predict $H_{\text{Eridanus}} \approx 75.9\ \text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ for the Eridanus supervoid—a parameter-free prediction falsifiable with current data. The framework further predicts a gravitational lensing Einstein radius of $\theta_E \approx 0.47$ arcsec for a Milky Way analog, distinguishable from ΛCDM lensing profiles. No new particles, fields, or free parameters are introduced beyond the single coupling constant ξ fixed by galaxy rotation curves.

Structural Zero Cosmological Constant and the Hubble Tension from Causal Actualization Density

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March 2026

1 Introduction

Modern cosmology confronts three problems that ΛCDM addresses by postulate rather than derivation.

The cosmological constant problem. Quantum field theory predicts a vacuum energy density $\rho_{\rm vac} \sim 10^{74}$ GeV^4; the observed value is $\rho_\Lambda \sim 10^{-47}$ GeV^4, a discrepancy of $\sim 10^{121}$ [1]. No dynamical mechanism selects this fine-tuned cancellation.

The Hubble tension. Early-universe measurements (CMB [2], BAO [3]) consistently return $H_0 \approx 67$–$68\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$; late-universe measurements (Cepheids [4], strong lensing time delays [5]) return $H_0 \approx 73$–$74\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$. The discrepancy stands at 4–6σ [6] and has not been resolved by revised systematic uncertainties.

The dark matter problem. Galaxy rotation curves [7], gravitational lensing, and large-scale structure all require a dominant non-luminous mass component. Yet no WIMP dark matter signal has been detected despite decades of direct searches [8, 9].

We propose that all three share a common structural cause: the metric is sourced not by mass-energy density but by *actualization density*—the rate at which irreversible causal interactions write permanent events into the growing spacetime graph. This reorientation follows from the Relational Actualism framework introduced in a companion paper [16], where it was shown that wave function collapse coincides with irreversible increase in quantum relative entropy $\Delta S(\rho\|\sigma_0) > 0$. That criterion implies: off-shell (virtual) processes produce no actualization events; causally silent particles produce no metric sourcing; and expansion is driven by the rate of vertex addition, which is density-dependent.

Section 2 presents the actualization field equations. Sections 3–5 derive the three cosmological consequences. Section 6 assembles four quantitative predictions. Section 7 positions the framework against ΛCDM and competing proposals.

2 The Actualization Density Field

2.1 Setup

Let spacetime be modelled as a growing causal directed acyclic graph (DAG) $G = (V, E)$, where each vertex $v \in V$ represents an actualization event and each directed edge $(u, v) \in E$ encodes the causal relation "u is in the causal past of v." The *actualization density* $\lambda(x)$ is the continuum limit of the vertex addition rate per unit spacetime volume at position x. In the dense-graph limit $\lambda \to \infty$, the DAG recovers a smooth Lorentzian manifold; for finite λ, discrete causal corrections appear.

The *coarse-grained assembly depth* $\tilde{A}_{\rm RA}(x)$ is the accumulated causal depth of the DAG up to x—the number of actualization events in the past light cone $J^-(x)$, weighted by their causal depth from x:

$$\tilde{A}_{\rm RA}(x) \;=\; \int_{J^-(x)} \lambda(x')\, d^4x'. \tag{1}$$

For matter in thermodynamic equilibrium over cosmological time, $\tilde{A}_{\rm RA}/V \propto \rho_{\rm matter}$, recovering the standard stress-energy source.

2.2 The metric sourcing equation

The weak-field metric sourcing equation in RA separates into two contributions that standard GR conflates:

$$\nabla^2\Phi \;=\; 4\pi G\big[\rho_A(x) + \rho_\lambda(x)\big], \tag{2}$$

with

$$\rho_A(x) = \frac{\hbar\langle\Delta E\rangle}{c^2} \cdot \frac{\tilde{A}_{\mathrm{RA}}(x)}{V}, \tag{3}$$

$$\rho_\lambda(x) = -\frac{\xi}{4\pi G}\,\nabla^2\big(\ln\lambda(x)\big). \tag{4}$$

The first term ρ_A is sourced by accumulated causal depth and recovers standard Newtonian gravity for equilibrium matter. The second term ρ_λ is a topological correction proportional to the Laplacian of the log actualization density; it vanishes when λ is spatially uniform and becomes significant at sharp density gradients.

Equation (2) is the weak-field limit of the covariant RA field equation:

$$G_{\mu\nu} \;=\; 8\pi G\Big(T^{(A)}_{\mu\nu} + \Theta^{(\lambda)}_{\mu\nu}\Big), \tag{5}$$

where $\Theta^{(\lambda)}_{\mu\nu} = \xi\big[\nabla_\mu\nabla_\nu(\ln\lambda) - g_{\mu\nu}\Box(\ln\lambda)\big]$ is the actualization tensor. In the static weak-field limit, $G_{00} \approx \nabla^2\Phi$ and $\Theta^{(\lambda)}_{00} \approx \xi\,\nabla^2(\ln\lambda)$, recovering equation (2).

2.3 Coupling constant

Dimensional analysis gives $\xi \sim r_d^2 v_{\mathrm{flat}}^2$, where r_d is the disk scale radius and v_{flat} is the asymptotic rotation velocity. Calibration against Milky Way analog rotation curves ($v_{\mathrm{flat}} = 220\ \mathrm{km\,s^{-1}}$, $r_d = 3$ kpc) gives

$$\frac{\xi}{GM_{\mathrm{disk}}} \;\approx\; 0.33, \tag{6}$$

consistent with natural order. This is the only free parameter of the framework; all subsequent predictions are derived from it or from observationally calibrated ratios.

3 Structural Zero Cosmological Constant

3.1 The vacuum does not gravitate

The core claim is:

Theorem 3.1 (Vacuum non-sourcing)**.** *The actualization contribution of the vacuum state to the metric source is exactly zero:*

$$\big\langle 0_M \;\big|\; \hat{T}^{\mu\nu}_{\mathrm{act}} \;\big|\; 0_M \big\rangle = 0. \tag{7}$$

Proof sketch. The metric source (5) is sourced by actualization events, which require $\Delta S(\rho\|\sigma_0) > 0$ (companion paper [16], criterion (3)). The Minkowski vacuum $|0_M\rangle$ is the reference state σ_0 itself; the relative entropy of σ_0 with respect to itself is identically zero. Consequently, no actualization event occurs in the vacuum and $\langle 0_M|\hat{T}^{\mu\nu}_{\mathrm{act}}|0_M\rangle = 0$. □

3.2 Why this is not fine-tuning

In ΛCDM, the $\sim 10^{121}$ discrepancy is absorbed by a fine-tuned cancellation between the bare cosmological constant and the vacuum energy contribution. No dynamical principle enforces or explains this cancellation.

In RA, equation (7) is a structural consequence of the actualization ontology: the metric is sourced by events that write vertices into the causal DAG, and the vacuum state, by definition, contains none. There is no large vacuum energy to cancel. The cosmological constant problem does not arise.

3.3 What drives expansion?

If vacuum energy does not drive the observed accelerated expansion, what does? The answer is the intrinsic graph growth mechanism (Section 4): matter and radiation interactions continuously add edges to the DAG, and the effective expansion rate is inversely proportional to the local vertex density. Regions of low actualization density (cosmic voids) expand faster than dense regions (galaxy clusters). The global average produces the observed cosmic acceleration without any Λ term.

3.4 CMB observational target

If vacuum energy contributes nothing to the metric source, the expansion history is governed entirely by matter and radiation actualization densities. The CMB power spectrum—particularly the low-multipole region where dark energy effects are most pronounced—should deviate from ΛCDM predictions. The angular power spectrum deficit at $\ell < 30$, which ΛCDM leaves unexplained, may be a signature of this mechanism. Quantifying the predicted deviation requires the full covariant extension (equation (5)), which is an active target of the programme.

4 The Hubble Tension as Actualization-Density-Dependent Expansion

4.1 The mechanism

The effective cosmic expansion rate satisfies:

$$H_{\text{eff}}(x) \;\propto\; \frac{1}{\lambda(x)}. \tag{8}$$

This follows from the bandwidth-partition structure of the causal DAG: the total actualization rate λ_{total} is divided between matter-binding edges (which produce the metric source ρ_A) and spatial-expansion edges (which drive cosmic expansion):

$$\lambda_{\text{total}} \;=\; \lambda_m + \lambda_{\text{exp}}. \tag{9}$$

High-λ regions (galaxy clusters, filaments) allocate most of their bandwidth to matter interactions; low-λ regions (intergalactic voids) allocate more bandwidth to expansion. The metric expands freely between sparse causal stitches in a way it cannot in a densely woven region.

4.2 The tension explained

Early-universe probes (CMB, BAO) sample the actualization density of the dense, radiation-dominated early universe, where λ is high. They measure H_0 in a high-λ regime.

Late-universe probes (Cepheids, Type Ia supernovae, strong lensing time delays) are dominated by lines of sight through void-dominated regions, where λ is lower. They measure H_0 in a low-λ regime.

Equation (8) predicts that these two families measure genuinely different physical quantities. The 4–6σ discrepancy is not a systematic error but a real signal of density-dependent expansion.

4.3 The bandwidth-partition formula

For a region with fractional density contrast $\delta = (\rho - \bar{\rho})/\bar{\rho}$, the local Hubble rate is:

$$\frac{H_{\text{local}}}{\bar{H}} \;=\; 1 - \delta\,\frac{\bar{\lambda}_m}{\bar{\lambda}_{\text{exp}}}, \tag{10}$$

where $\bar{\lambda}_m/\bar{\lambda}_{\text{exp}}$ is the ratio of the global mean matter-binding actualization rate to the mean expansion rate.

Calibration from the KBC void. The KBC void [10] has density contrast $\delta \approx -0.20$ and radius $\sim$ 300 Mpc. Using SH0ES [4] $H_{\text{local}} = 73.04\ \text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ and Planck [2] $\bar{H} = 67.4\ \text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$:

$$\frac{\bar{\lambda}_m}{\bar{\lambda}_{\text{exp}}} \;=\; \frac{H_{\text{local}}/\bar{H} - 1}{-\delta} \;=\; \frac{0.084}{0.20} \;\approx\; 0.42. \tag{11}$$

This is a one-parameter calibration.

4.4 Falsifiable prediction: the Eridanus supervoid

The Eridanus supervoid has density contrast $\delta \approx -0.30$ and diameter $\sim$300 Mpc [11]. Substituting into equation (10) with the calibrated ratio (11):

$$H_{\text{Eridanus}} \;=\; 67.4\times\big(1-(-0.30)(0.42)\big) \;\approx\; 75.9\ \text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}. \tag{12}$$

This prediction is *parameter-free given the calibration*: it uses no additional inputs beyond the KBC fit and the measured density contrast of Eridanus.

This is structurally distinct from ΛCDM (which predicts H_0 to be a global constant) and from standard void models (which produce at most 1–2 $\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ corrections from peculiar velocities [10]). A measured value of H in the direction of the Eridanus supervoid consistent with $\approx 75.9\ \text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ would strongly support the bandwidth-partition mechanism.

4.5 Line-of-sight dependence

A second discriminating prediction follows directly from equation (10): the inferred H_0 from any cosmological probe should correlate with the mean matter density along its line of sight. Void-dominated survey geometries should return systematically higher H_0; cluster-dominated geometries should return lower values. This line-of-sight dependence is not predicted by ΛCDM or by early dark energy models.

5 WIMP Prohibition and the Dark Matter Signature

5.1 The WIMP Prohibition

A particle species that undergoes no on-shell electromagnetic or strong-force interactions produces no actualization events and therefore no contribution to ρ_A. The actualization rate for a canonical 100 GeV WIMP in the Milky Way ISM

($n_b \sim 1$ nucleon cm^{-3}, $v \sim 200$ km s^{-1}) with spin-independent cross-section $\sigma_\chi \lesssim 10^{-44}$ cm^2 is [8]:

$$\Gamma_\chi \;=\; n_b\,\sigma_\chi\,v \;\approx\; 2\times 10^{-37}\ \mathrm{s}^{-1} \text{ per WIMP.} \quad (13)$$

The mean time between actualization events for a WIMP is $\tau = 1/\Gamma_\chi \approx 5\times 10^{36}$ s, which is $\sim 10^{18}$ times the age of the universe. A WIMP halo of five times the baryonic mass contributes $\lesssim 10^{-25}$ of the metric sourcing of the visible matter. This is a *categorical prohibition*: WIMPs cannot account for galactic rotation curves or gravitational lensing within RA, regardless of their abundance.

This prediction is consistent with the null results of LZ [8], PandaX-4T [9], and XENONnT—which have pushed WIMP-nucleon cross-sections to 10^{-48} cm^2—and provides a structural rather than observational reason for the absence.

5.2 The $\nabla^2(\ln\lambda)$ rotation curve mechanism

The dark matter phenomenology is explained by the ρ_λ term in equation (2). At the edge of a galactic disk where the baryonic density falls off, $\nabla^2(\ln\lambda) < 0$ (actualization is diverging into the sparse halo), producing a gravitational contribution that persists beyond the disk.

For a galaxy with exponential baryonic surface density $\Sigma(r) = \Sigma_0 e^{-r/r_d}$, the ρ_λ correction generates an additional circular velocity:

$$v_\lambda^2(r) \;\approx\; \xi\,\frac{d}{dr}\left[r\,\frac{d}{dr}\ln\lambda\right], \quad (14)$$

which at large r (where $\lambda \propto r^{-\alpha}$ follows the baryonic profile) produces a flat asymptotic contribution $v_\lambda^2(r) \to \xi\,\alpha^2/r^0 = \text{const}$. The total rotation curve

$$v^2(r) \;=\; v_{\mathrm{bary}}^2(r) + v_\lambda^2(r) \quad (15)$$

becomes asymptotically flat without a dark matter halo.

The Tully-Fisher scaling $v_{\mathrm{flat}}^4 \propto M_{\mathrm{bary}}$ emerges naturally from equation (15) in the limit $r \gg r_d$, recovering MOND's most successful phenomenological prediction from a first-principles actualization mechanism.

5.3 Gravitational lensing and the Bullet Cluster

Gravitational lensing in RA traces the actualization density distribution $\lambda(x)$ rather than total mass. The Einstein radius for a galaxy cluster of velocity dispersion σ_v at redshift z_L lensing a source at z_S is:

$$\theta_E \;=\; \sqrt{\frac{4G\,M_{\mathrm{act}}}{c^2}\cdot\frac{D_{LS}}{D_L D_S}}, \quad (16)$$

where M_{act} is the total actualization-weighted mass (the ρ_A contribution within the Einstein radius). For a Milky Way analog ($v_{\mathrm{flat}} = 220$ km s^{-1}), equation (16) gives

$$\theta_E \;\approx\; 0.47 \text{ arcsec}, \quad (17)$$

which is measurably smaller than the ΛCDM prediction of ~ 0.9–1.4 arcsec for a comparable halo mass.

The Bullet Cluster. The Bullet Cluster offset between the lensing centroid and the X-ray gas [12] is fully consistent with RA. The RA explanation does not invoke dark matter: the lensing traces the baryonic distribution of the merged cluster *before the gas has time to re-equilibrate*—a transient in which ρ_A (which tracks the accumulated causal depth of the stars, which passed through the merger without colliding) exceeds ρ_A of the gas (which shock-heated and decelerated). The lensing centroid follows the stars, not the gas, because the stars have higher accumulated assembly depth $\tilde{A}_{\mathrm{RA}}$.

6 Summary of Predictions

Table 1 summarizes the four quantitative predictions of the framework. All four are parameter-free given the single coupling ξ calibrated in Section 2.3, or the $\bar{\lambda}_m/\bar{\lambda}_{\mathrm{exp}}$ ratio calibrated from the KBC void.

7 Discussion

7.1 Comparison to ΛCDM

ΛCDM resolves the accelerated expansion by inserting a cosmological constant by hand, introduces a dark matter particle by assumption, and has no dynamical explanation for the Hubble tension (which it treats as a systematic error). The present framework resolves all three from a single additional structural ingredient: the metric is sourced by actualization density, not mass-energy density. The cosmological constant problem does not arise because the vacuum does not gravitate. The Hubble tension has a specific geometric mechanism. Dark

Table 1: Quantitative predictions of the causal actualization density framework. All four predictions are structurally distinct from ΛCDM (denoted by − in the ΛCDM column). AE = Actualization explanation; ✓ = prediction made by framework; × = prediction absent.

Prediction	Value	ΛCDM	RA	Test
H in Eridanus supervoid (δ=−0.30)	$75.9\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$	global const.	✓	Ongoing surveys [11]
H_0 vs. line-of-sight mass density	anticorrelation	no dep.	✓	DESI, Euclid void catalog
WIMP metric sourcing suppression	$\leq 10^{-25}$ of baryons	mass prop.	✓	LZ, PandaX-4T null results
MW-analog Einstein radius	≈ 0.47 arcsec	0.9–1.4 arcsec	✓	Strong lensing surveys

matter phenomenology is reproduced by the ρ_λ term.

7.2 Comparison to MOND and Verlinde

Modified Newtonian Dynamics (MOND) [13] introduces a critical acceleration a_0 by hand. Verlinde's emergent gravity [14] derives a MOND-like effect from an entropic argument but does not give a Hubble-tension mechanism or resolve the cosmological constant. The ρ_λ term in equation (2) recovers MOND's phenomenological success—specifically, the Tully-Fisher relation $v_{\mathrm{flat}}^4 \propto M_{\mathrm{bary}}$—from a first-principles causal mechanism, without introducing a_0 as a free parameter. The value of a_0 emerges as a derived consequence of ξ and the mean cosmological actualization density.

7.3 Comparison to early dark energy

Early dark energy (EDE) models [15] attempt to resolve the Hubble tension by raising H_0 uniformly via an additional energy component in the pre-recombination universe. The bandwidth-partition mechanism makes a structurally different prediction: H_0 is not raised uniformly but *varies as a function of the local actualization density.* The line-of-sight dependence of H_0 (Section 4) is the key discriminating signature. DESI's void catalog and Euclid's weak-lensing maps will provide the data needed to test this prediction directly.

7.4 Open mathematical targets

The framework has two acknowledged open derivations.

Hard Wall 1: The Bianchi compatibility condition. The RA field equation (5) must satisfy the contracted Bianchi identity $\nabla_\mu G^{\mu\nu} \equiv 0$, which requires $\nabla_\mu \Theta^{\mu\nu}_{(\lambda)} = 0$. In the weak-field static limit this is satisfied by construction; the full covariant proof requires deriving ξ from the microscopic graph dynamics.

Hard Wall 2: The covariant $\Box(\ln\lambda)$ target. The weak-field $\nabla^2(\ln\lambda)$ of equation (4) must be elevated to the covariant $g^{\mu\nu}\nabla_\mu\nabla_\nu(\ln\lambda)$ to be applicable in strong-field and cosmological regimes. These targets do not undermine the weak-field predictions presented here, which are derived in the regime where the approximations hold.

8 Conclusion

We have shown that the cosmological constant problem, the Hubble tension, and the dark matter problem share a common structural origin. When the spacetime metric is sourced by actualization density $\lambda(x)$ rather than mass-energy density, three consequences follow without additional parameters: the vacuum does not gravitate (Theorem 3.1); the Hubble rate is density-dependent, producing a real physical tension between early- and late-universe probes; and causally silent particles are categorically prohibited from gravitating, while baryonic rotation-curve phenomenology is recovered from the ρ_λ term.

The four quantitative predictions of Table 1 are all testable with current or near-future surveys. The Eridanus prediction (12) is immediately falsifiable with existing void survey data. The Einstein radius prediction (17) is testable with current strong-lensing surveys. The line-of-sight H_0 dependence is testable with DESI and Euclid.

These results flow from the same irreversibility criterion developed in the companion measurement

paper [16]. The connection to the quantum foundations is direct: off-shell processes produce no actualization events (they satisfy $\Delta S = 0$), which is why the vacuum does not gravitate. The same physical principle that dissolves the measurement problem also dissolves the cosmological constant problem.

Acknowledgements

The author acknowledges the use of Anthropic's Claude and Google's Gemini as AI research partners during the preparation of this manuscript. The author is solely responsible for all scientific content and conclusions.

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